

CRISTINA CUESTA BRET

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Early-career Mechanical Engineer with hands-on experience in mechanical design and manufacturing. Passionate about renewable/sustainable technologies, innovating in mechanical design, and applying cutting-edge solutions to enhance product efficiency and user experience.

EDUCATION

Master's Programme in Environomical Pathways for Sustainable Energy Systems

Sept. 2026 – May 2028

InnoEnergy Masters+ – TU Eindhoven & KTH Royal Institute of Technology

Bachelor of Science in Mechanical Engineering

GPA: 3.6

Certificate in Artificial Intelligence Fundamentals and Applications

University of Florida, Gainesville

Aug. 2021 – May 2026

Relevant Coursework: Thermodynamics, Fluid Mechanics, Energy Conversion, Controls, Materials, FEA, Manufacturing, Engineering Leadership

SKILLS

Software Skills: SolidWorks, MATLAB, Python, Microsoft Office, Granta, PDM, Abaqus, MS Excel

Technical Skills: 3D CAD software, manufacturing engineering, FEA, programming, technical writing, engineering drawings, materials

Languages: Spanish (Native), English (Fluent)

WORK EXPERIENCE

Teaching Assistant

Aug. 2023 – May 2026

University of Florida

Gainesville, FL

- **Dynamics and Controls Systems Design Laboratory (Fall 2025).** Supervised 6 weekly labs using a Furuta Pendulum; provided feedback on transfer functions, bang-bang, PID, and full-state feedback control as well as LabVIEW coding. Graded 50+ lab reports.
- **Mechanics of Materials Laboratory (Spring 2025).** Led 4 weekly labs, mentoring students on tensile testing, strain gauges, pressure experiments, and LabVIEW coding; assessed 60+ lab reports with detailed feedback.
- **Elements of Electrical Engineering (Fall 2023 – Fall 2024).** Reinforced fundamental electrical engineering concepts and circuit building.

Mechanical Engineering Summer Intern

Jul. 2025 – Aug. 2025

GMV Aerospace and Defence SAU

Madrid, Spain

- Collaborated with senior engineers on the CAD and assembly of electronic hardware systems for defense and aerospace applications.
- Independently executed a full design project using SolidWorks, collaborating with the production team to adapt to evolving constraints and deliver a manufacturable solution.
- Operated and maintained 3D resin printers to validate and refine prototypes through iterative assessments.
- Supported structural, vibration, and shock analyses on 2 mechanical components to ensure reliability under operational loads.
- Executed material selection analyses for 2 projects, balancing mechanical performance and application constraints.

Undergraduate Research Intern

May 2023 – July 2023

Universitat de Lleida

Lleida, Spain

- Developed a sophisticated 700-line solar management Python algorithm aimed at optimizing the allocation of Direct Normal Irradiance (DNI) resources and providing accurate power output forecast based on dynamic demands and boundary conditions.
- Streamlined Python code to support various receiver sizes and electrical configurations of photovoltaic (PV) cells, improving the tool's versatility and performance by 20%.

INVOLVEMENT

Chassis and Cockpit Engineer

Jan. 2023 – May 2026

Solar Gators

Gainesville, FL

- Designed and manufactured the steering wheel for the solar car, implemented LCD and paddle shifters for regenerative braking.
- Executed multiple hands-on manufacturing tasks, including carbon fiber wet layups, to support vehicle production.
- Designed and manufactured an actuator release body latch system and canopy hinge using SolidWorks contributing to the structural and functional integrity of the car while following FSGP and ASC regulations.

President, Secretary and Social Media Manager

May. 2022 – May 2026

University of Florida Club Tennis

Gainesville, FL

- **First female** to lead the University of Florida Club Tennis team with +150 members.
- Manage home and away tournaments with +30 teams.
- Directed the design of team uniforms and increased sales by 60%. Managed social media accounts with +2,500 followers, increasing engagement and impressions by 80%.

Honors: Herbert Werheim College of Engineering **Dean's List**, **Deputy Lead** in Senior Mechanical Design Capstone, **Honors Thesis**

Certifications: SolidWorks, CNC Machining Training Program, Cambridge Proficiency in English (CPE)